

Course notes for

# Algebraic Topology I (KSM3E02)

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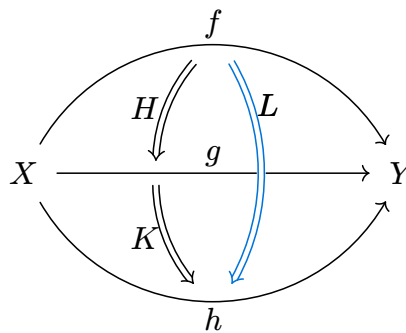
**Day 2 : 5<sup>th</sup> August, 2026**

homotopy extension – homotopy lifting

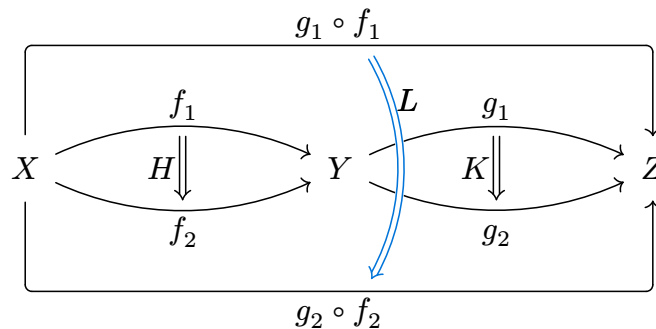
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## 2.1 Composition and Homotopies

There are two ways to compose homotopies. We can compose them *vertically*



This is nothing but the transitivity. We can also compose them *horizontally*



### Exercise 2.1:

Construct the horizontal composition of two homotopies.

## 2.2 Retraction

**Definition 2.2:** (*Retraction*)

Given a subspace  $A \subset X$ , a **retraction** is a map  $r : X \rightarrow A$  such that  $r \circ \iota = \text{Id}_A$ , where  $\iota : A \hookrightarrow X$  is the inclusion. We say  $A$  is a **retract** of  $X$ . In other words, we have a commuting diagram

$$\begin{array}{ccc} & r & \\ A & \xleftarrow{\quad} & X \\ & \iota & \end{array}$$

As an example, any point in a space is a retract. We shall see that the boundary circle is *not* a retract of the closed disc (and more generally, boundary sphere is not a retract of the closed ball). Try to justify that  $A = \{0, 1\} \subset [0, 1]$  is *not* a retract of  $[0, 1]$ !

**Exercise 2.3:**

Suppose  $X$  is a  $T_2$ -space (i.e, Hausdorff). If  $A \subset X$  is a retract, verify that  $A$  must be a closed subset.

**Hint :** For a  $T_2$ -space  $X$ , if we have two maps  $f, g : X \rightarrow Y$ , then the diagonal  $\Delta = \{x \in X \mid f(x) = g(x)\}$  is a closed subset.

Let us now specialize homotopy equivalence to the case of retraction, which is easier to visualize.

**Definition 2.4:** (*Deformation Retract*)

A subspace  $A \subset X$  is a **deformation retract** if there is a retraction  $r : X \rightarrow A$  satisfying  $r \circ \iota = \text{Id}_A$ , and there is a homotopy  $\iota \circ r \simeq \text{Id}_X$ . Explicitly, we need a homotopy  $H : X \times [0, 1] \rightarrow X$  such that

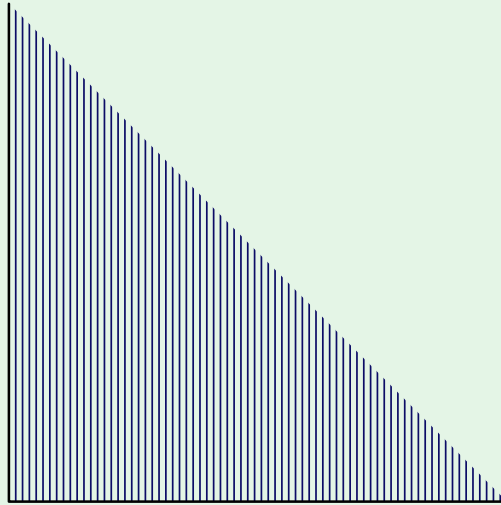
$$H(x, 1) = x, \quad x \in X, \quad H(x, 0) \in A, \quad x \in X, \quad H(a, 0) = a, \quad a \in A.$$

We say  $A \subset X$  is a **strong deformation retract** if moreover the homotopy restricts to identity on  $A$ , i.e, if  $H(a, t) = a$  for all  $a \in A$  and  $0 \leq t \leq 1$ .

It is clear from the definition that a deformation retract is always a homotopy equivalence. There are (pathological) examples which are deformation retracts, but *not* strongly.

**Example 2.5:**

Consider the following comb-like space.



Explicitly, the space  $X$  can be identified as the following subspace of  $\mathbb{R}^2$

$$X = ([0, 1] \times \{0\}) \cup \bigcup_{r \in [0,1] \cap \mathbb{Q}} \{r\} \times [0, 1 - r]$$

That is, keep the horizontal interval  $[0, 1] \times \{0\}$ , and then for each rational point  $r$  on it add the vertical line of length  $1 - r$ .

We claim that  $X$  strongly deformation retracts onto any point on the horizontal line. Indeed, it is easy to construct a linear homotopy which pushes everything down to the horizontal line, and then collapses the line to a designated point; this designated point is never moved throughout.

On the other hand, for any point *not* on the horizontal line, there is always a deformation retract: just perform the previous deformation to the point, say,  $(0, 0)$  and then slide this point to any other point, speeding up as necessary. But for such a point, there cannot be a strong deformation retract. To see this intuitively, observe that if there is a strong deformation retract to a point, then in any neighborhood of that point must contain a smaller neighborhood which also collapses to that same point. But this is clearly not the case since you can always find a neighborhood of such a point which cannot be path connected (and hence cannot collapse to a point).

Let us give a name to spaces which are homotopy equivalent to a point.

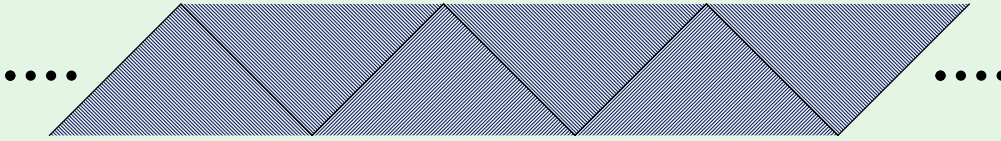
**Definition 2.6:** (*Contractible Space*)

A space  $X$  is said to be **contractible** if it is homotopy equivalent to a point. We denote it as  $X \simeq \star$ .

We have seen that any space admits a retraction to any point in it. It is easy to see that if a space deformation retracts to a point, then it is contractible, and vice versa. This does not hold true for *strong* deformation retract.

### Example 2.7:

This example builds upon [Example 2.5](#). Consider the following subspace  $X$  of  $\mathbb{R}^2$



This is obtained from the comb-like space in [Example 2.5](#) by arranging them in the infinite zigzag pattern!

We claim that this space  $X$  is *contractible*, but does not *strongly* deformation retract to any point. Intuitively, a strong deformation retract to a point  $x_0$  will imply that there for any neighborhood of  $x_0$ , there is a smaller neighborhood which contracts to  $x_0$ : this is clearly not the case here (do a case-by-case analysis of different kinds of points).

In order to see why this space is contractible, we do the following. Firstly, for each point  $x$  in the space, we have a unit-speed path  $r_x : [0, \infty) \rightarrow X$  which follows the *bristle* of the comb where  $x$  belongs till it hits the bold zigzag line, and then moves along the zigzag to the right. Clearly if  $x$  is already on the zigzag, it just starts following it. Consider the following map

$$F : X \times [0, 1] \rightarrow X$$

$$(x, t) \mapsto r_x(t).$$

One can check that  $F$  is a continuous map, and hence a homotopy between the  $\text{Id}_X$  to a map which ends in the zigzag. Finally, the zigzag is just a copy of  $\mathbb{R}$ , and hence contractible. Thus, composing we can get a contracting homotopy. In other words,  $X \simeq \star$ .

### Caution 2.8:

According to Hatcher, a “deformation retract” is precisely what we call a *strong* deformation retract. Hatcher also defines a notion of “deformation retract in the weak sense”: this asks for a deformation retraction  $H : X \times [0, 1] \rightarrow X$  such that  $H(a, t) \in A$  for all  $a \in A$  and for all time  $0 \leq t \leq 1$ .

## 2.3 Homotopy Extension Property

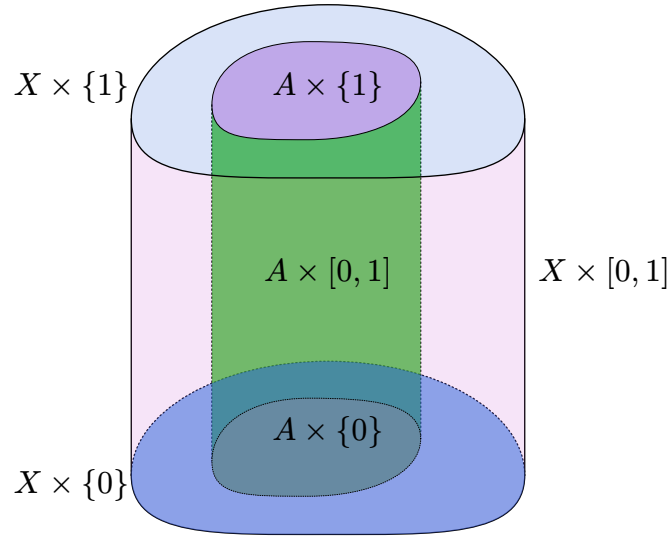
Suppose  $A \subset X$  is a subspace. Assume that we are given a map  $f : X \rightarrow Y$ , and a homotopy  $h : A \times [0, 1] \rightarrow Y$  from  $f|_A : A \rightarrow Y$ . We say, the pair  $(X, A)$  has *homotopy extension property* (or *HEP*) with respect to the space  $Y$  if there exists a homotopy  $H : X \times [0, 1] \rightarrow Y$  from  $f$ , which extends the homotopy  $h$ . That is,

$$H(x, 0) = f(x) \quad x \in X, \quad H(a, t) = h(a, t) \quad a \in A, 0 \leq t \leq 1.$$

The schematic picture explains what is going on. We are given a continuous map on the subspace

$$A \times [0, 1] \cup X \times \{0\} \subset X \times [0, 1],$$

and we wish to extend it to the whole space  $X \times [0, 1]$ .



Homotopy extension for  $A \subset X$ .

**Definition 2.9: (Cofibration)**

The inclusion map  $\iota : A \hookrightarrow X$  is called a *(Hurewicz) cofibration* if it has homotopy extension property with respect to every space  $Y$ .

**Remark 2.10:**

More generally, we can define HEP for any map  $f : A \rightarrow X$ , not just inclusion maps. Indeed, this can be formulated via the following diagram

$$\begin{array}{ccc}
 A & \xrightarrow{h} & Y^{[0,1]} \\
 f \downarrow & \nearrow H & \downarrow e \\
 X & \xrightarrow{g} & Y
 \end{array}$$

Here,  $Y^{[0,1]}$  denotes the space of continuous maps  $[0, 1] \rightarrow Y$ , and  $e : Y^{[0,1]} \rightarrow Y$  is evaluation at 0. A homotopy  $h : A \times [0, 1] \rightarrow Y$  can be interpreted as a map  $h : A \rightarrow Y^{[0,1]}$ . The map  $f : A \rightarrow X$  is said to have HEP against the space  $Y$  if for any given map  $g : X \rightarrow Y$  and a homotopy  $h : A \times [0, 1] \rightarrow Y$ , such that  $h|_{A \times \{0\}} = g \circ f$ , there exists a homotopy  $H : X \times [0, 1] \rightarrow Y$  which makes the diagram commutative. One then defines  $f : A \rightarrow X$  to be a *(Hurewicz) cofibration* if it has HEP against all space  $Y$ .

Turns out any cofibration  $f : A \rightarrow X$  is actually injective, and moreover,  $f$  is a homeomorphism onto its image. Thus, we are not losing any generality with our definition.

Let us now see a prototypical example of a cofibration!

**Example 2.11:** ( $S^{n-1} \hookrightarrow D^n$  is a Cofibration)

Consider the inclusion  $\iota : S^{n-1} \hookrightarrow D^n$  of the unit  $(n-1)$ -sphere  $S^{n-1}$  as the boundary of the closed unit disc  $D^n$  in  $\mathbb{R}^n$ . We claim that this has HEP against any space, and hence, is a cofibration. Suppose  $f : D^n \rightarrow Y$  is a map, and  $h : S^{n-1} \times [0, 1] \rightarrow Y$  is a homotopy of  $f|_{S^{n-1}}$ . Consider the subspace

$$X := S^{n-1} \times [0, 1] \cup D^n \times \{0\} \subset D^n \times [0, 1].$$

Clearly, we are given a map  $F : X \rightarrow Y$ , which we intend to extend to  $D^n \times [0, 1]$ . One can imagine  $X$  as a hollowed out cylinder (with the bottom intact, basically like a bucket). One can define a *retraction*  $r : D^n \times [0, 1] \rightarrow X$  as follows. Consider the point  $z = \mathbf{0} \times \{1\}$ , the center of the disc  $D^n \times \{1\}$ . Given any point  $p$  in the solid cylinder  $D^n \times [0, 1]$ , draw a straight line from  $z$  which passes through  $p$ , and then record the (unique) point  $q$  where this line hits the bucket  $X$ . Define  $r : D^n \times [0, 1] \rightarrow X$  which maps  $p \mapsto q$ . It is easy to see that  $r$  is continuous, and moreover,  $r$  is identity when restricted to the subspace  $X$ . Define the homotopy  $D^n \times [0, 1] \rightarrow Y$  as the composition  $D^n \times [0, 1] \xrightarrow{r} X \xrightarrow{F} Y$ . Since  $r$  is identity on  $X$ , this composition is indeed an extension of  $F$ . Thus,  $\iota : S^{n-1} \hookrightarrow D^n$  is a cofibration.

**Exercise 2.12:**

The construction of retraction map in [Example 2.11](#) is also known as the *orthographic projection*. Figure out an explicit formula for the map!